GyaanVault

SPPU M.Sc. Data Science — Free Study Material

Machine Learning

DS-552-MJ

Interview Preparation — Topic-wise Q&A

Semester II · Core

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Curated interview questions and answers for Machine Learning, organised by topic from the SPPU M.Sc. Data Science syllabus. Ideal for viva, placement and internship preparation.

Tip: read the answer, then close the PDF and try to explain each concept aloud in your own words.

Fundamentals

Q. Supervised vs unsupervised learning?

Supervised learning uses labelled data to predict outputs (classification/regression); unsupervised learning finds structure in unlabelled data (clustering, association).

Q. What is overfitting and how to prevent it?

When a model learns noise and performs poorly on new data. Prevent via cross-validation, regularization (L1/L2), more data, or simpler models.

Q. Explain the bias-variance tradeoff.

High bias underfits (too simple); high variance overfits (too complex). Good models balance both to minimise total error.

Q. What is regularization?

Adding a penalty on coefficient size — Lasso (L1) can zero-out features; Ridge (L2) shrinks them; ElasticNet combines both.

Algorithms & evaluation

Q. How does SVM work?

It finds the hyperplane that maximally separates classes; kernels map data to higher dimensions for non-linear separation.

Q. How does KNN classify a point?

By majority vote of its k nearest neighbours using a distance metric; simple but sensitive to k and feature scaling.

Q. Explain precision, recall and F1-score.

Precision = TP/(TP+FP), recall = TP/(TP+FN), F1 = harmonic mean of the two. Used when classes are imbalanced.

Q. What is a confusion matrix?

A table of TP, TN, FP, FN counts used to derive accuracy, precision, recall and other classification metrics.

Q. K-Means vs hierarchical clustering?

K-Means partitions into k clusters by minimising within-cluster distance (needs k in advance); hierarchical builds a dendrogram tree without pre-specifying k.